

Unsupervised Learning - Lab 10 - Bayesian Networks

Exact and Approximate Inference

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Bayesian Networks (BN)

- A **Bayesian Network** is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a Directed Acyclic Graph (DAG).
- Nodes represent variables and edges represent direct dependencies.
- Each node is associated with a probability function that takes, as input, a particular set of values for the node's parent variables.

Conditional Probability Distributions (CPD)

- A **Conditional Probability Distribution** defines the probability of a variable given its parents in the graph.
- For a variable X with parents $Pa(X)$, the CPD is $P(X|Pa(X))$.
- CPDs are essential for defining the joint:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | Pa(X_i))$$

- **Inference** in Bayesian Networks involves computing the posterior probability distribution of a set of query variables given some evidence.
- Two main types of inference:
 1. **Exact Inference**: Computes the exact posterior probabilities. Methods include variable elimination and the junction tree algorithm.
 2. **Approximate Inference**: Provides an approximation of the posterior probabilities. Methods include sampling techniques like Gibbs sampling and variational inference.

- Exact inference guarantees correct results but can be computationally expensive.
- **Variable Elimination:** Systematically eliminates variables from the network to compute the marginal probability.
- **Junction Tree Algorithm:** Transforms the network into a tree structure to facilitate efficient computation.

Exact Inference

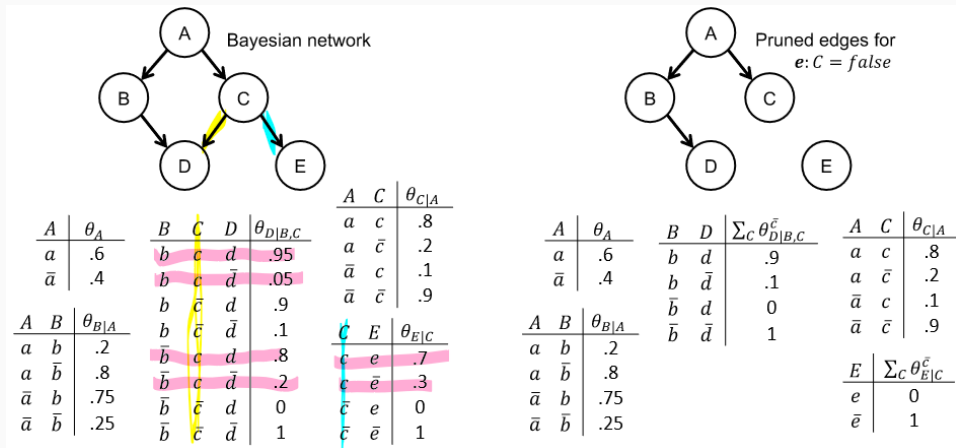


Figure 1: Example of Exact Inference

Approximate Inference

- Approximate inference methods are used when exact methods are infeasible.
- **Sampling Methods:** Generate samples from the distribution to approximate the posterior. Examples include Gibbs sampling and importance sampling.
- **Variational Inference:** Optimizes a family of distributions to approximate the posterior.

Approximate Inference

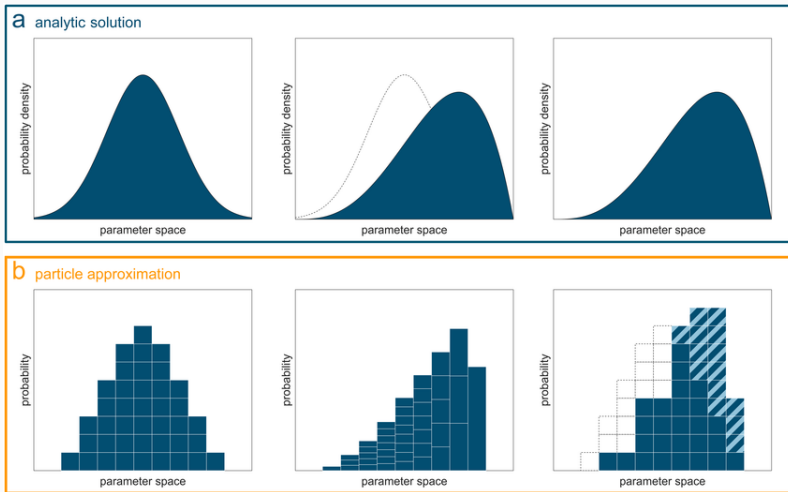


Figure 2: Example of Approximate Inference

- Understanding CPDs is crucial for defining the joint probability distribution.
- Inference methods, both exact and approximate, enable the computation of posterior probabilities.
- Choice of inference method depends on the trade-off between accuracy and computational feasibility.

- pgmpy documentation: pgmpy.org
- Alarm network BIF file: [BN Repository](#)